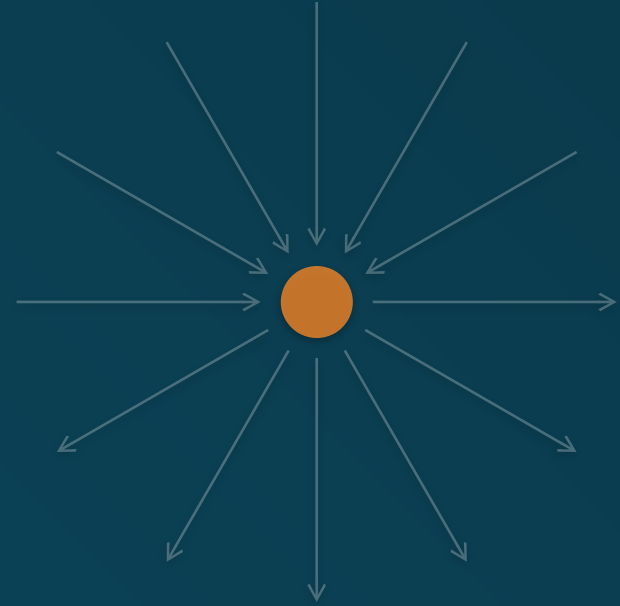


TOPIC 20

Magnetic Fields

磁场



CIE 9702 · A Level Physics A2 · Paper 4

剑桥国际物理 A2 分章节讲义 · 中英双语

Learning Outcomes

学习目标

LEARNING OUTCOMES

学习目标

By the end of this topic, you will be able to...

20.1–20.2 Field & force on a conductor 磁场与安培力

- Magnetic field as a field of force; field lines 磁场概念与磁感线
- $F = BIL \sin\theta$ with Fleming's left-hand rule 左手定则
- Define magnetic flux density B (tesla) 磁通密度定义

20.3 Force on a moving charge 洛伦兹力

- $F = BQv \sin\theta$; circular motion in a uniform field 圆周运动
- Hall voltage $V_H = BI/(ntq)$ and the Hall probe 霍尔效应

20.4 Fields due to currents 电流的磁场

- Field patterns of a straight wire, flat coil and solenoid 三种载流体的磁场分布

20.5 Electromagnetic induction 电磁感应

- Magnetic flux $\Phi = BA$ and flux linkage $N\Phi$ 磁通量与磁链
- Faraday's law and Lenz's law 法拉第定律与楞次定律

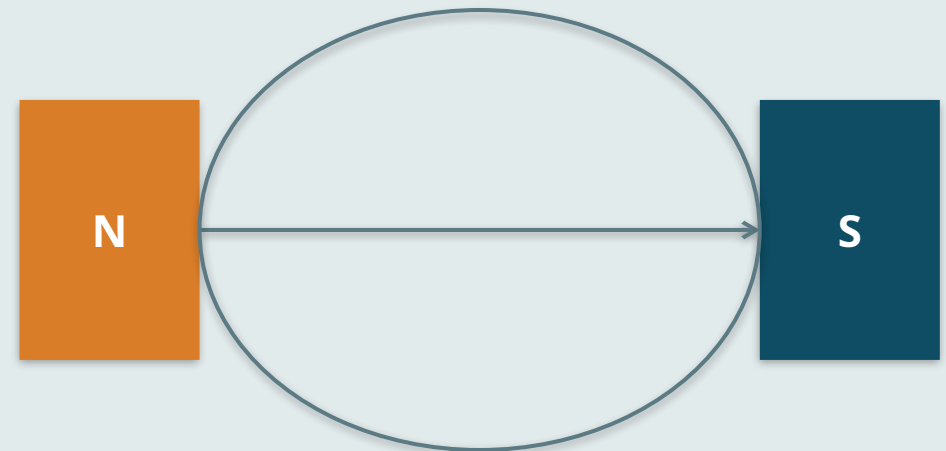
Concept of a Magnetic Field

磁场的概念

A **magnetic field** is a region of space where a force is experienced by a **magnetic pole**, a **moving charge** or a **current-carrying conductor**.

磁体、运动电荷或载流导体受力的区域。

- Produced by **permanent magnets** or by **moving charges (currents)** 永磁体或电流产生
- Field lines: $N \rightarrow S$ outside the magnet 外部 N 到 S ; density shows strength 疏密表强弱
- Oersted (1820): a current deflects a compass — currents create magnetic fields 奥斯特发现电流磁效应



Field lines of a bar magnet 条形磁铁的磁感线

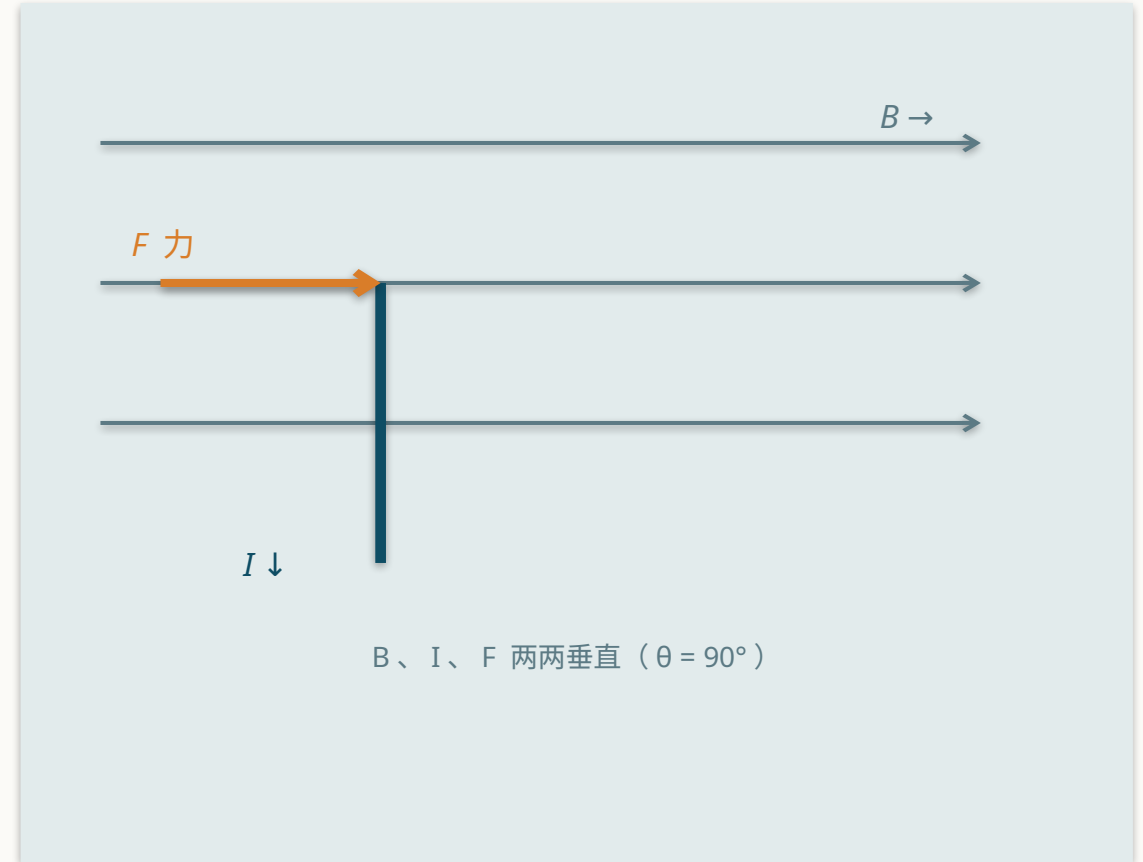
Force on a Current-Carrying Conductor

通电导线的受力

$$F = B I L \sin \theta$$

$\theta =$ 电流与磁场的夹角

- Maximum force when wire $\perp$ field ($\theta = 90^\circ$) ; zero when parallel 垂直最大，平行为零
- Direction: **Fleming's left-hand rule** 左手定则
- First finger **F**ield, se**C**ond finger **C**urrent, thu**M**b **M**otion 手指定则口诀



Magnetic Flux Density

磁通密度 B 的定义

Magnetic flux density B: the force per unit current per unit length on a wire placed **at right angles** to the magnetic field.

垂直于磁场的导线，单位电流单位长度所受的力。

$$B = \frac{F}{IL} \quad (\theta = 90^\circ)$$

单位 tesla (T) = N A⁻¹ m⁻¹

- B is a **vector**: direction of the field line tangent 矢量
- A **current balance** measures B from $F = BIL$ 电流天平测 B
- Typical values: Earth's field $\approx 10^{-5}$ T ; laboratory magnets 0.01–1 T 量级参考

Force on a Moving Charge

运动电荷的受力

- Current is moving charge: $\Rightarrow$ single charge feels 由导线受力争得

$$F = BQv \sin \theta$$

洛伦兹力

- Direction: left-hand rule on the **conventional current** direction 左手定则按电流方向
- **Negative charges move opposite to current** — force direction reverses 负电荷反向!
- Force is always $\perp$ velocity $\Rightarrow$ changes direction, not speed 只改变方向不改变速率



Circular Motion in Magnetic Fields

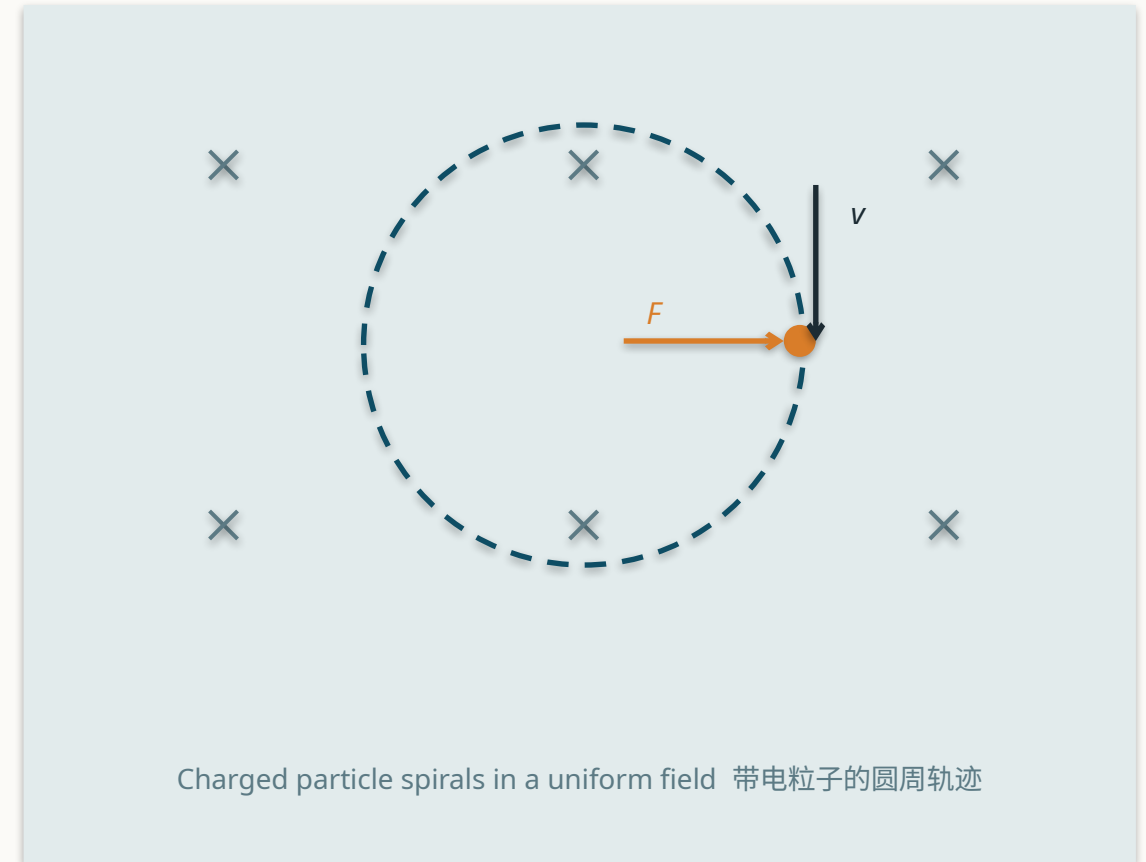
匀强磁场中的圆周运动

- $F = BQv$ is always $\perp v \Rightarrow$ provides the **centripetal force** 洛伦兹力充当向心力

- $\Rightarrow$

$$r = \frac{mv}{BQ} \quad T = \frac{2\pi m}{BQ} \quad \text{半径与周期}$$

- Period is **independent of speed** 周期与速率无关
- Applications 应用: mass spectrometer 质谱仪, cyclotron 回旋加速器, fine-beam tube 电子比荷测定



The Hall Effect

霍尔效应

- Charge carriers are deflected by the magnetic force and accumulate on one face $\Rightarrow$ transverse **Hall voltage** V_H 载流子偏转积累形成横向电压
- Equilibrium: magnetic force = electric force $\Rightarrow$ 平衡条件
- With :

$$V_H = \frac{BI}{n t q}$$

n 载流子密度 $\cdot t$ 厚度

- Metals: huge $n \Rightarrow$ tiny V_H (μV) 金属霍尔电压极小
- Semiconductors: small $n \Rightarrow$ measurable V_H 半导体明显 $\Rightarrow$ 实用探头材料



Hall slice 霍尔片：载流子偏转形成横向电压

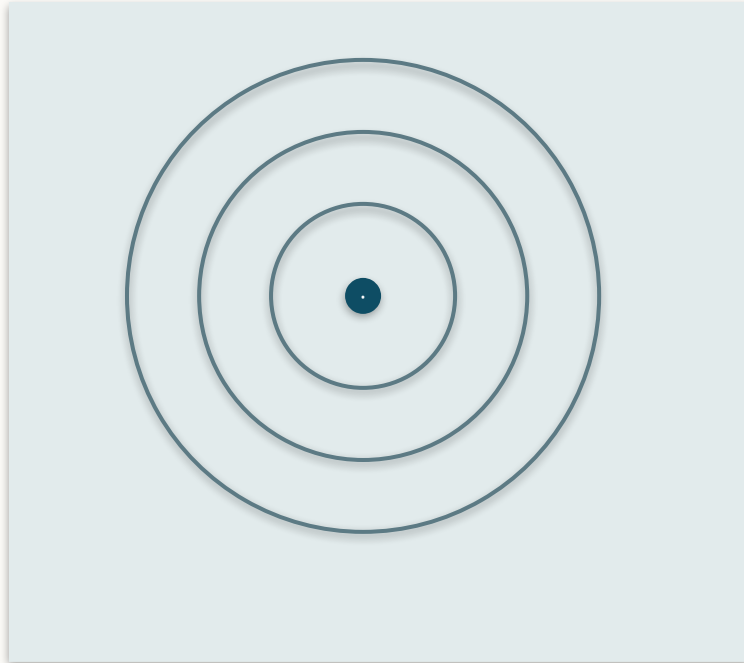
The Hall Probe

霍尔探头测磁场

- (for fixed I , n , t , q) $\Rightarrow$ the voltage **measures** flux density 电压正比于 B
- Calibrate against a known field, then read B directly 校准后直读
- Measures only the component of B **normal to the probe slice** 只测垂直分量
- Small and fast: used to map field distributions (e.g. inside a solenoid) 可测磁场分布
- Angle θ to the field: 角度修正

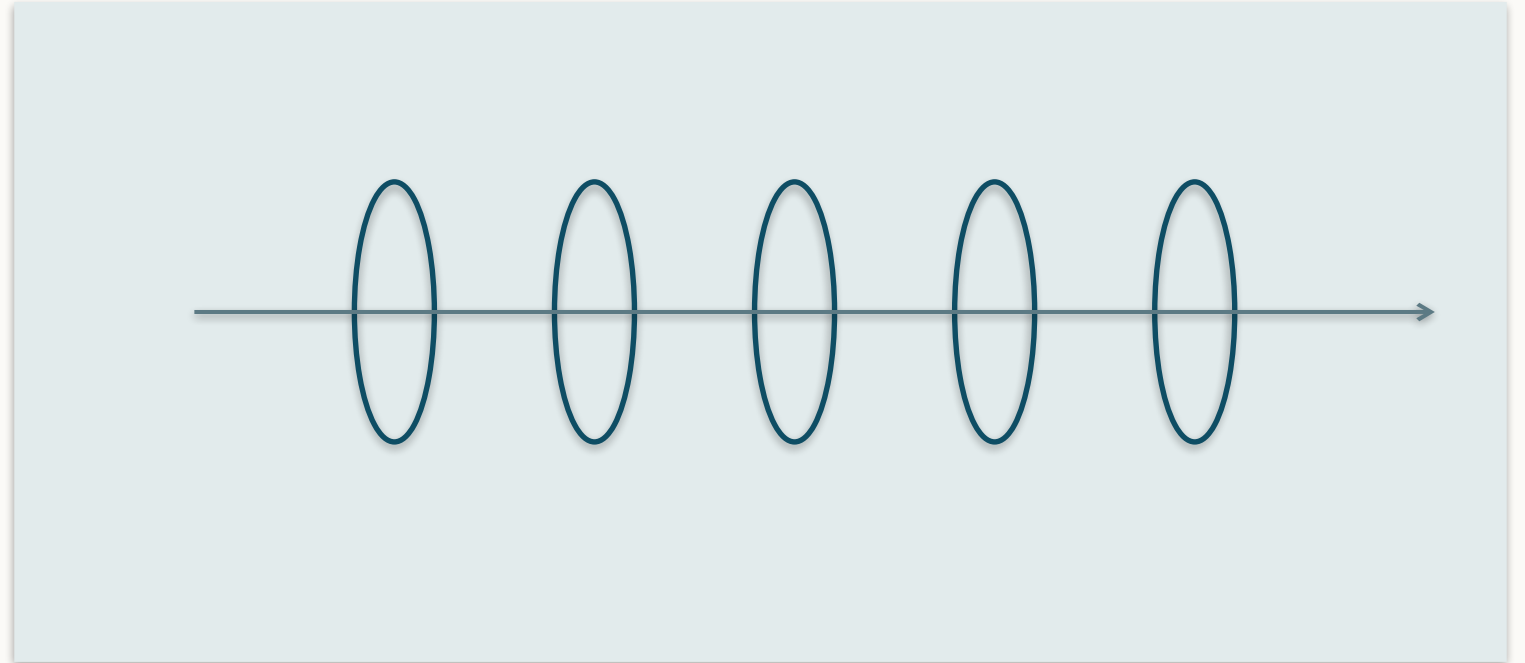
Fields due to Currents

电流产生的磁场



Long straight wire 长直导线:

Right-hand grip rule 右手螺旋定则: 拇指指电流, 四指指磁场



Solenoid 螺线管: 内部近似匀强 (n = 单位长度匝数)

Field like a bar magnet outside; uniform inside 外部类似条形磁铁

Flat circular coil 圆线圈: field strongest at the centre, ; $\mu_0 = 4\pi \times 10^{-7} \text{ H m}^{-1}$

Forces between Currents

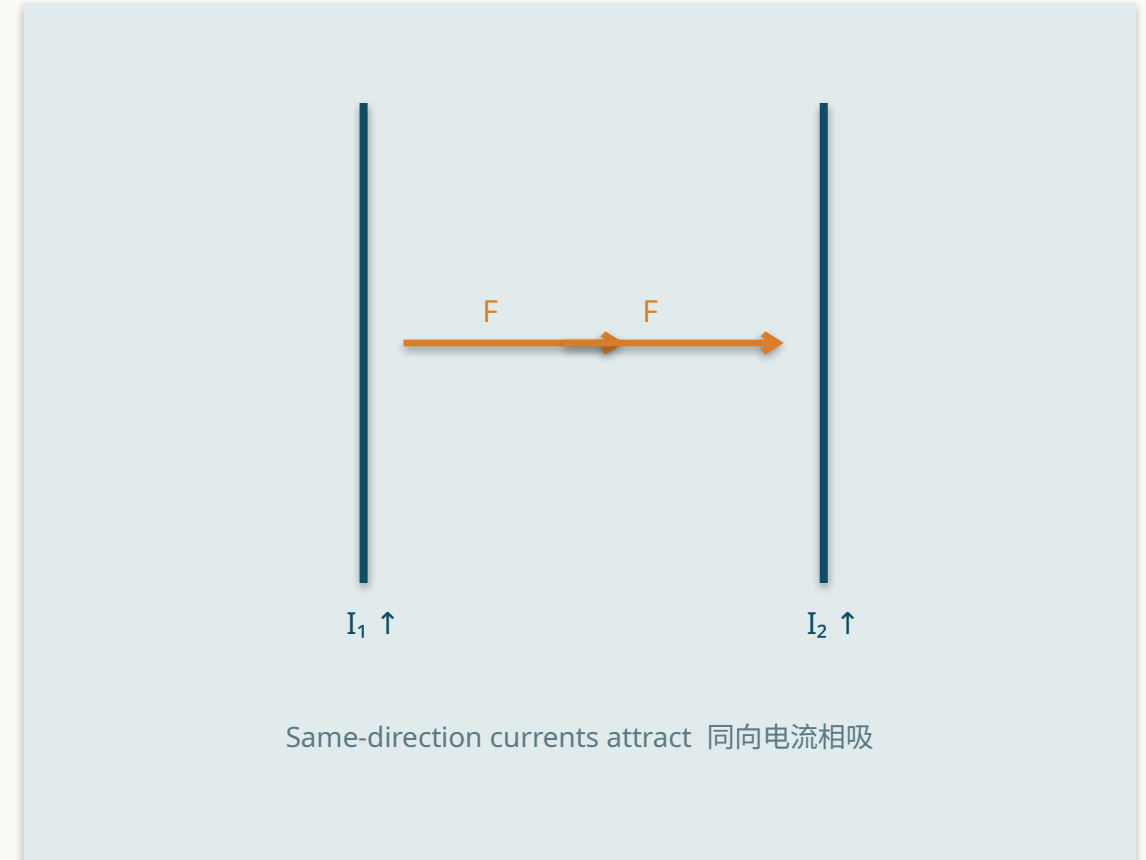
平行载流导线间的作用力

- Each wire sits in the field of the other: 一条导线处在另一条导线的磁场中

$$\frac{F}{L} = \frac{\mu_0 I_1 I_2}{2 \pi d}$$

单位长度的作用力

- **Same-direction currents attract; opposite repel** 同向相吸，反向相斥
- This defines the **ampere** 安培的定义基础



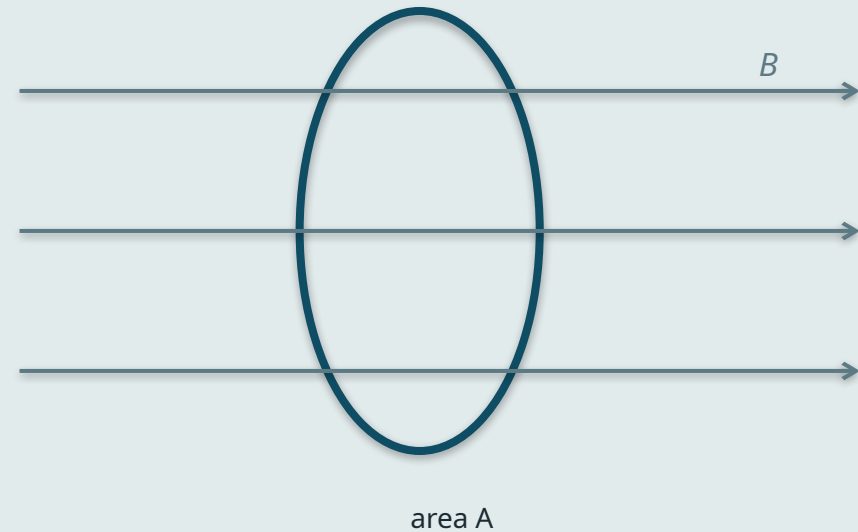
Magnetic Flux & Flux Linkage

磁通量与磁链

Magnetic flux : the product of flux density and the area **normal to the field**. Unit: weber (Wb).

垂直于磁场方向的面积与磁通密度的乘积。

- For a coil of N turns: **flux linkage** 磁链
- Field at angle θ to the normal: 斜交修正
- B can be read as "flux per unit area" $\Rightarrow$ 磁通密度名称由来



Flux through a coil 穿过线圈的磁通量

Faraday's Law

法拉第电磁感应定律

The induced e.m.f. is proportional to the **rate of change of flux linkage**.

感应电动势正比于磁链的变化率。

$$e = - \frac{d(N\Phi)}{dt}$$

负号见楞次定律

- Flux can change via: **B** changing, **area A** changing, or **angle** changing 三种变化途径
- Moving a magnet into a coil, or a wire cutting field lines, induces an e.m.f. 磁铁插入线圈 / 导线切割磁感线
- Magnitude only matters: faster change $\Rightarrow$ bigger e.m.f. 变化越快电动势越大

Lenz's Law

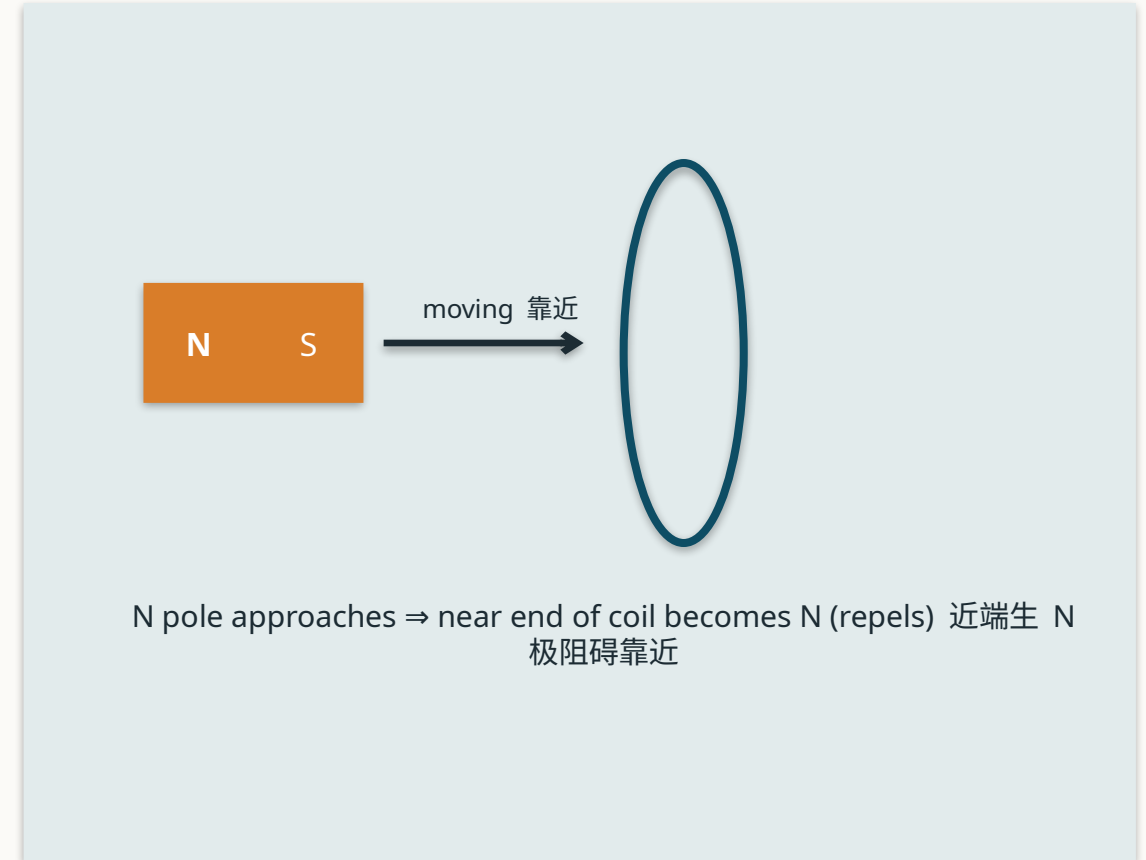
楞次定律

The direction of the induced e.m.f. (current) is such as to **oppose the change** producing it.

感应电流的效果总是阻碍引起感应的变化。

1. Identify the original flux direction 判断原磁通方向
2. Is the flux increasing or decreasing? 判断磁通增减
3. Induced field opposes the change 感应磁场阻碍该变化
4. Use the right-hand grip rule to get the current direction 右手定电流方向

• Lenz's law is **conservation of energy**: work must be done against the opposing force 楞次定律 = 能量守恒



Applications of Electromagnetic Induction

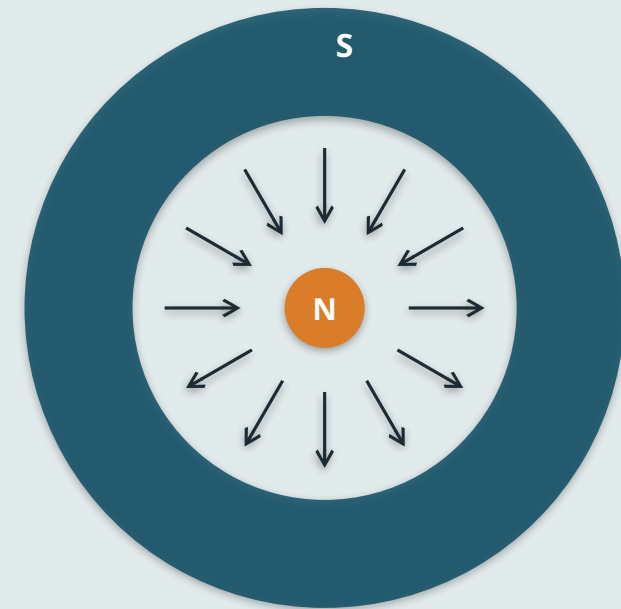
电磁感应的应用

- **A.C. generator** 交流发电机 : rotating coil in a field $\Rightarrow$ flux linkage varies sinusoidally $\Rightarrow$
- **Electromagnetic braking** 电磁制动 : eddy currents oppose motion (Lenz) 涡流制动
- **Induction charging / cooking** 感应充电 / 电磁炉 : changing field induces currents in a device or pan
- Microphones, credit-card stripes, metal detectors 话筒 / 探测器等

Worked Example 20A

例题解析：磁场作图

- **Question 题目** : A circular magnet has its north pole at the centre, separated from the surrounding circular south pole by an air gap. **Draw the magnetic field lines in the gap.** 圆环磁铁中心为 N 极、外环为 S 极，画出气隙中的磁感线。
- **Method 方法** : field lines run from N to S 磁感线由 N 指向 S
- In the gap they are **radial**, pointing outwards from the central N 气隙中呈放射状向外
- Evenly spaced $\Rightarrow$ uniform strength in the gap 等间距表示气隙中近似匀强



Radial field lines in the air gap 气隙中的放射状磁感线

Worked Example 20B

例题解析

WORKED EXAMPLE 20B

The horizontal component of the Earth's magnetic flux density is . The current in a horizontal cable is 120 A. Calculate (a) the maximum and (b) the minimum force per unit length, stating the angle in each case.

θ 是电流与磁场的夹角，注意不是与法线的夹角。

SOLUTION 解答

Step 1 Force per unit length: .

Step 2 (a) Maximum at : .

Step 3 (b) Minimum when the cable is **parallel** to the field: .

Answer (a) at 90° (b) zero (parallel 平行时)

Worked Example 20C

例题解析

WORKED EXAMPLE 20C

Electrons are accelerated to and enter a uniform field of flux density 0.50 mT. Their path is a circle of radius 10.2 cm. Calculate (a) the specific charge of the electron; (b) the mass of the electron ().

这就是历史上测电子比荷（荷质比）的方法。

SOLUTION 解答

Step 1 Circular motion: $\Rightarrow$.

Step 2 (a) .

Step 3 (b)

Answer (a) (b)

Worked Examples 20D & 20E

例题解析

WORKED EXAMPLE 20D

A velocity selector must select ions of speed v . The electric field strength is E . Calculate the magnetic flux density required.

1. Undeviated $\Rightarrow \Rightarrow$
- 2.

Answer

WORKED EXAMPLE 20E

A Hall probe at right angles to a uniform field reads 82 mV. The probe is then turned so its plane makes 35° with the field. Calculate the new Hall voltage.

1. Only the normal component counts:
- 2.

Answer

Class Practice

随堂练习

- Q1** Electrons are accelerated through 250 V, then enter a uniform field of 0.58 mT at right angles. Calculate (a) their speed ($m = 9.1 \times 10^{-31}$ kg, $e = 1.6 \times 10^{-19}$ C); (b) the radius of the circular path.
- Q2** Singly charged ions pass undeviated through a velocity selector with $E = 4.2 \times 10^4$ V m $^{-1}$ and $B = 8.4 \times 10^{-3}$ T. Calculate the selected speed. 速度选择器。
- Q3** Aluminium foil 0.10 mm thick carries 4.5 A; $n = 6.0 \times 10^{28}$ m $^{-3}$; a field of 85 mT is normal to the foil. Calculate the Hall voltage. 霍尔电压计算。

Answers Q1 (a) (b) 9.2 cm Q2 Q3 (0.40 μ V)

Summary

本章小结

KEY FORMULAS 核心公式

$$F = B I L \sin \theta$$

force on wire 安培力

$$F = B Q v \sin \theta$$

force on charge 洛伦兹力

$$r = \frac{m v}{B Q}$$

circular path 半径

$$V_H = \frac{B I}{n t q}$$

Hall voltage 霍尔电压

$$B = \frac{\mu_0 I}{2 \pi r}$$

straight wire 直导线

$$\Phi = B A$$

flux 磁通量

$$e = - \frac{d(N \Phi)}{d t}$$

Faraday 法拉第

$$\frac{F}{L} = \frac{\mu_0 I_1 I_2}{2 \pi d}$$

two wires 平行导线

KEY IDEAS 核心思想

- Left-hand rule for forces; right-hand grip rule for fields 左力右场。
- Magnetic force changes direction, never speed 洛伦兹力不做功。
- Faraday gives magnitude, Lenz gives direction 法拉第定大小，楞次定方向。

Exam Tips

备考提示

- Left-hand rule uses **conventional current**: reverse the direction for electrons/negative charges. 负电荷受力反向。
- Hall voltage derivation steps: force balance → substitute → . 推导按步给分。
- Flux **linkage** = $N\Phi$ — do not forget the N turns in Faraday's law. 磁链别漏乘 N 。
- Period of circular motion is independent of v — a favourite short question. 周期与速率无关。
- For Lenz's law, state: change → opposing induced field → current direction. 三步表述完整。